Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

Ruifeng Huang and Zhenglong Cui

Abstract—Light field depth estimation is crucial for 3D vision but remains highly challenging in mixed-material scenes. Existing epipolar plane image (EPI)-based methods heavily rely on the Lambertian assumption, suffering severe performance degradation when physical assumptions fail on non-Lambertian (e.g., specular, scattering) and complex-textured surfaces. To address this, a Unified Dual-Mask Physical Model (GeometricDualMask) was developed to explicitly decouple material properties from depth cues. Specifically, a three-layer angular signal decomposition model was introduced to physically disentangle the angular signal into illumination, disparity, and material residual components. Based on extracted geometric features, a dual-branch architecture adaptively routed features to dedicated Lambertian EPI and non-Lambertian geometric branches via a pixel-wise material mask. The model was optimized using a staged training strategy with domain-balanced sampling and guided training utilizing angular gradient pseudo-labels for precise mask learning. Extensive experiments demonstrate that the proposed method achieves an overall Mean Absolute Error (MAE) of 0.133 and a remarkably low MAE of 0.081 on mixed urban scenes. Furthermore, through systematic experiments, the physical boundaries of EPI methods are rigorously defined, revealing that the coupling of data scarcity and physical assumption violation fundamentally bottlenecks non-Lambertian performance. Ultimately, this work provides a robust physical decoupling framework for complex light field depth estimation and establishes critical physical boundaries to guide future research directions.

Index Terms—Light field depth estimation, Non-Lambertian surfaces, Epipolar plane image, Angular signal decomposition, Dual-branch network, Physical boundary analysis

I. Introduction

THE main contributions of this paper are summarized as follows:

Light field (LF) imaging captures both spatial and angular information of light rays, enabling advanced 3D perception tasks in autonomous driving, virtual reality, and multi-view video systems . Depth estimation serves as a core component in these systems, providing essential geometric priors for downstream applications such as view synthesis, scene reconstruction, and video rendering. Among various LF depth estimation paradigms, Epipolar Plane Image (EPI) based methods leverage the linear structure of EPIs to extract disparity, achieving high accuracy in ideal environments .

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However, the efficacy of EPI-based methods relies heavily on the Lambertian reflectance assumption, which posits that surface appearance remains consistent across different viewpoints, mathematically forming linear structures in EPIs as $I(x,u) \approx f(x+u\cdot d)$, where d denotes disparity. This assumption fails in real-world scenarios containing non-Lambertian surfaces (e.g., specular reflections, translucent materials) and complex textures. In non-Lambertian regions, viewpoint-dependent radiance shifts destroy the linear EPI structure, while complex textures cause severe slope ambiguity even under Lambertian conditions. The severity of this degradation is evident in empirical evaluations. When applied to mixed urban scenes (e.g., UrbanLF-Syn) and synthetic datasets (e.g., HCInew), conventional EPI networks yield a Mean Absolute Error (MAE) of 0.411 in non-Lambertian regions, significantly exceeding the acceptable threshold of 0.25. Similarly, the MAE in complex-textured Lambertian regions reaches 0.387, far above the 0.16 target. Extensive empirical investigations involving 132 experiments across 16 architectural and optimization directions—including attention mechanisms, dual-branch networks, and composite loss functions—demonstrate that merely adjusting network topologies or hyperparameters cannot overcome these physical violations. This performance bottleneck is further exacerbated by extreme data scarcity, as existing non-Lambertian LF datasets provide only four training scenes, creating a hard barrier for data-driven generalization.

To address these physical and data-induced limitations, we propose a Unified Dual-Mask Physical Model for non-Lambertian LF depth estimation. Instead of relying solely on end-to-end feature learning, our approach explicitly decouples material properties from geometric disparities at the physical level. We introduce a three-layer angular signal decomposition model that formulates the LF angular signal as $I(u,v) = I_{dc} + a\cdot u + b\cdot v + \varepsilon(u,v)$, separating it into a direct current component I_{dc} (environmental illumination), a linear component (disparity/depth defined by slopes a and b), and a residual component $\varepsilon(u,v)$ (material/reflectance properties). Based on this decomposition, a GeometricDualMask architecture is designed to generate material masks and route features adaptively to specialized Lambertian and non-Lambertian processing branches. Domain-balanced sampling is incorporated to maintain exposure equilibrium across mixed scenes during joint training.

The main contributions of this work are summarized as follows:

- We propose a three-layer angular signal decomposition model and a geometric feature-based material decoupling mechanism. By extracting geometric features such as angular gradients and correlation coefficients, this mechanism achieves high-precision binary classification of Lambertian and non-Lambertian regions, providing a theoretical framework to understand EPI failure causes.
- We develop a GeometricDualMask dual-branch architecture integrated with domain-balanced sampling. This design utilizes the generated material masks to adaptively route features, effectively mitigating the performance degradation of single-EPI networks in mixed-material scenes and improving depth estimation accuracy in complex urban environments.
- We systematically define the physical boundaries of EPI assumption failures through extensive empirical analysis. By evaluating 16 major research directions, we reveal the coupling effect of physical assumption violations and data scarcity, demonstrating that pure EPI frameworks possess inherent architectural limitations in non-Lambertian and complex-textured scenes, thereby providing critical guidance for future non-EPI or neural radiance field-based research.

The remainder of this paper is organized as follows. Section II reviews related work in LF depth estimation and non-Lambertian surface processing. Section III details the three-layer angular signal decomposition and the geometric feature extraction mechanism. Section IV presents the GeometricDualMask architecture and the domain-balanced training strategy. Section V provides comprehensive experimental results, cross-dataset evaluations, ablation studies, and computational cost analyses. Finally, Section VI concludes the paper.

Traditional light field depth estimation methods primarily rely on handcrafted features and photo-consistency metrics, such as variance and gradient-based operators applied to Epipolar Plane Images (EPIs) . These approaches operate under the strict Lambertian reflectance assumption. The primary limitation lies in their rigid adherence to this physical model. In the presence of non-Lambertian surfaces, such as specular reflections or translucent materials, viewpoint-dependent radiance shifts violate the photo-consistency constraint, causing the linear EPI structure to collapse. Consequently, these methods yield erroneous disparity estimates in highlight and scattering regions. At the technical level, handcrafted features lack the capacity to distinguish between geometric disparity and material-induced radiance variations. In complex-textured Lambertian regions, high-frequency spatial details introduce severe slope ambiguity in EPIs, leading to matching errors and depth discontinuities that traditional regularizations cannot effectively resolve.

To mitigate the reliance on handcrafted features, early deep learning-based methods, such as EPINet and its variants, utilize convolutional neural networks to implicitly learn disparity from EPI stacks [1]. While these models improve accuracy in ideal scenarios, they inherit and sometimes amplify the physical limitations of the EPI representation. Methodologically, these networks treat depth estimation as a pure regression task without explicit material awareness. They employ a single, unified processing pipeline for all pixels, forcing the network to fit a single EPI slope distribution across heterogeneous materials. In mixed urban scenes, this uniform treatment causes significant performance degradation, as the network struggles to balance the distinct geometric characteristics of Lambertian and non-Lambertian regions. Technically, early networks extract features directly from the raw angular signals, entangling environmental illumination, geometric parallax, and surface reflectance. When confronted with non-Lambertian regions, the network attempts to regress depth from corrupted EPI lines, resulting in severe depth artifacts and elevated Mean Absolute Error (MAE), often exceeding 0.40 in non-Lambertian evaluations.

Recent advancements have introduced complex architectural modifications, including attention mechanisms, multi-scale feature fusion, dual-branch structures, and defocus-based auxiliary heads, aiming to enhance robustness in challenging scenarios . Despite these architectural innovations, they encounter intrinsic physical and data-driven bottlenecks. The core methodological flaw is the attempt to overcome physical assumption failures through sheer architectural complexity. Extensive empirical evaluations involving 132 experiments across 16 optimization directions reveal that simply increasing network capacity or adjusting loss functions cannot bypass the physical limits of EPI-based geometry. When the angular cone geometry constraint is violated by non-Lambertian reflectance, attention weighting or feature recalibration cannot reconstruct the missing linear EPI structure. From a technical perspective, these advanced models suffer from the coupling effect of physical assumption violation and extreme data scarcity. Non-Lambertian light field datasets are exceptionally limited, often containing only a few training scenes. Complex models with numerous parameters easily overfit to these scarce samples, failing to generalize to unseen material properties. They lack a physical signal decomposition mechanism to explicitly decouple the DC component (illumination), linear component (parallax), and residual component (material), thereby wasting representational capacity on modeling illumination variations rather than geometric depth.

The systematic analysis of these limitations indicates that purely data-driven architectural tweaks are insufficient to resolve depth estimation degradation in mixed-material scenes. Overcoming specific error thresholds, such as reducing the non-Lambertian MAE below 0.25 and the complex-textured Lambertian MAE below 0.16, requires a paradigm shift from implicit feature learning to explicit physical and geometric decoupling. Therefore, there is an urgent need for a method that explicitly models the physical boundaries of EPI assumptions, decouples material properties from geometric signals at the angular level, and adaptively routes features based on material masks to achieve robust depth estimation in complex, mixed-material light field scenes.

In this paper, we propose a Unified Dual-Mask Physical Model, termed GeometricDualMask, for light field depth estimation that explicitly addresses the physical limitations of Epipolar Plane Images (EPIs) in mixed-material scenes. Rather than employing a single regression pipeline, our approach decomposes the angular signal into illumination, disparity, and material components. Guided by this physical decomposition, a dual-branch architecture adaptively routes features to specialized processing streams for Lambertian and non-Lambertian regions, supervised by a pixel-level material mask.

- Three-Layer Angular Signal Decomposition and Geometric Material Decoupling: We formulate a physical decomposition model $I(u, v) = DC + a \cdot u + b \cdot v + \varepsilon(u, v)$ that separates the light field angular signal into illumination (DC), disparity (linear coefficients), and material reflectance (residuals). To overcome the coarse frequency resolution under limited angular sampling (e.g., 9×9), we extract geometric features, such as angular gradients and correlation coefficients, instead of frequency-domain features. This mechanism achieves high-precision binary classification of Lambertian and non-Lambertian materials, providing a theoretical framework to analyze EPI degradation on non-Lambertian surfaces.
- Dual-Mask Geometric Routing Architecture for Mixed Scenes: We design a dual-branch network that utilizes the extracted geometric features to generate a pixel-level material mask, adaptively routing features to a Lambertian EPI branch and a Non-Lambertian geometric branch. The Lambertian branch leverages continuous depth priors to preserve EPI slope structures, while the non-Lambertian branch captures local geometric anomalies, such as cross-shaped patterns in EPIs. Incorporating domain-balanced sampling and weighted random sampling, this architecture mitigates the performance degradation of unified EPI networks in complex urban and mixed-material scenarios.
- Root Cause Analysis and Physical Boundary Delineation of EPI Assumptions: Through an extensive empirical study comprising 132 experimental iterations across 16 architectural and training directions, we systematically delineate the physical boundaries of EPI-based depth estimation. We demonstrate that the coupling of data scarcity and the violation of the angular cone constraint inherently limits EPI performance on specular and scattering surfaces. This analysis provides critical negative findings, indicating that overcoming non-Lambertian challenges requires moving beyond pure EPI representations toward neural radiance fields or learned feature matching with large-scale datasets.

Extensive evaluations on the HCInew, UrbanLF-Syn, and non-Lambertian datasets demonstrate that the proposed method achieves an overall Mean Absolute Error (MAE) of 0.133 and a mixed urban MAE of 0.081. While the architecture yields significant accuracy improvements in overall and mixed domains, the systematic evaluation concurrently validates the theoretical physical limits of EPI representations in highly textured and non-Lambertian regions.

II. Related Work

A. Traditional Light Field Depth Estimation Methods

Early light field depth estimation methods primarily extract geometric cues from the physical properties of light rays, with Epipolar Plane Image (EPI) slope analysis being a dominant approach. Wanner et al. formulate depth estimation as an orientation estimation problem in the EPI space. By computing the structure tensor of EPI lines, this method extracts local disparity values based on the dominant gradient direction. A continuous global optimization framework incorporating Markov Random Fields (MRF) is subsequently employed to enforce spatial smoothness and resolve depth ambiguities. This approach demonstrates high accuracy in regions with distinct textures and clear linear EPI structures, establishing a foundational baseline for EPI-based geometry extraction.

While EPI slope analysis performs effectively in textured regions, it degrades significantly in textureless areas where gradient information is sparse. To address this limitation, Tao et al. [2] integrate defocus cues with correspondence cues to construct a more robust depth estimation pipeline. The correspondence cue relies on photo-consistency across different viewpoints, similar to traditional stereo matching, whereas the defocus cue measures the sharpness of refocused images at varying depth planes. By combining these two complementary physical signals through a probabilistic framework, the method yields reliable depth maps even on weakly textured surfaces, expanding the applicability of light field depth estimation beyond purely texture-dependent EPI slope calculations.

Advancing the multi-view geometric perspective, Jeon et al. tackle the challenge of low angular resolution in commercial light field cameras, which restricts the baseline for conventional stereo matching. They introduce a subpixel phase shift technique to align multi-view images accurately, enabling the computation of high-precision correspondence costs. By incorporating adaptive support weights and handling occlusions explicitly, this method enhances depth estimation in regions with fine details and narrow structures. The integration of phase-shifted multiview stereo with light field data effectively mitigates the quantization errors inherent in discrete angular sampling.

Despite their theoretical elegance, these traditional methods share significant limitations rooted in their reliance on handcrafted features and the strict Lambertian reflectance assumption. First, they assume that surface radiance remains invariant across viewpoints, mathematically forming linear EPI structures as $I(x, u) \approx f(x + u \cdot d)$.

When scenes contain non-Lambertian materials, such as specular reflections or translucent surfaces, viewpoint-dependent radiance shifts destroy the linear EPI topology, causing the structure tensor and photo-consistency metrics to fail entirely [3]. Second, handcrafted features like gradients and structure tensors blur in complex texture regions and lack the capacity to explicitly decouple material properties from geometric depth. Consequently, these methods cannot distinguish whether an EPI distortion originates from a depth discontinuity or a surface reflection. Finally, existing approaches process mixed-material scenes uniformly, lacking a systematic root-cause analysis and physical boundary definition for EPI assumption failures. Unlike these traditional frameworks that suffer from severe performance degradation in non-Lambertian regions, our work introduces a three-layer angular signal decomposition model to decouple geometric features from material properties. By explicitly defining the physical boundaries of EPI validity and employing a dual-mask routing architecture, the proposed method adaptively handles mixed-material scenes, preventing the ineffective computational waste associated with applying pure EPI frameworks to non-Lambertian surfaces.

B. Deep Learning-based General Light Field Depth Estimation Methods

Building upon the physical cue extraction discussed in the previous section, deep learning-based methods have shifted the paradigm from handcrafted geometric descriptors to data-driven feature learning. Early convolutional neural network (CNN) architectures primarily focus on extracting disparity cues directly from Epipolar Plane Images (EPIs). For instance, Shin et al. [1] propose a multi-scale CNN that projects the 4D light field into 2D EPI slices to regress depth values. By learning the mapping between EPI slope patterns and disparity, this approach achieves high accuracy in textured Lambertian regions. However, the single-branch architecture implicitly assumes a uniform reflectance model, making the learned EPI features highly sensitive to viewpoint-dependent radiance variations caused by non-Lambertian surfaces.

To compensate for the limited receptive field of local EPI patches, subsequent research integrates sub-aperture images (SAIs) and macropixel features to capture broader spatial-angular correlations. Heber et al. [2] introduce a framework that jointly processes EPIs and macropixels, fusing local geometric slopes with global spatial context. More recently, Transformer-based architectures have been adopted to model long-range dependencies within the light field. The Light Field Transformer (LFT) [4] utilizes epipolar-shifted window attention to aggregate features across distant viewpoints, effectively reducing depth ambiguity in textureless areas. Despite these architectural advancements, the feature fusion remains implicit. The network treats all angular signals uniformly, failing to distinguish between the linear structures of Lambertian regions and the distorted patterns of specular or translucent materials.

Recognizing the vulnerability of standard EPI features to complex textures and reflectance, several methods incorporate attention mechanisms and adaptive loss functions to enhance robustness. For example, hierarchical attention networks [2] assign dynamic weights to different angular views, attempting to suppress inconsistent viewpoints that violate the photo-consistency assumption. Similarly, confidence map estimation approaches [2] down-weight unreliable disparity predictions in occluded or highly reflective areas. While these strategies yield moderate improvements in controlled environments, they rely on statistical regularization rather than physical modeling. The network is forced to learn a compromised representation that averages the characteristics of both Lambertian and non-Lambertian signals within a single feature space.

Despite the continuous evolution from basic CNNs to complex Transformers, these general light field depth estimation methods share significant limitations when applied to real-world mixed scenes. First, single-branch architectures typically assume the scene consists of pure Lambertian surfaces or simple mixtures, leading to significant performance degradation in complex urban environments where Lambertian and non-Lambertian materials coexist. Second, these methods lack explicit perception and feature routing mechanisms for non-Lambertian regions, causing the network to produce severe depth artifacts in specular highlights and fail to achieve material-adaptive processing. Third, implicit feature learning cannot explain the physical root causes of EPI failure. Empirical evidence indicates that merely increasing network complexity, introducing attention modules, or adjusting loss functions cannot overcome the physical limits imposed by the violation of the EPI assumption. These inherent bottlenecks necessitate a paradigm shift from implicit single-branch learning to explicit physical modeling. This motivates our unified dual-mask physical model, which decouples material properties via geometric features to enable a dual-mask routing architecture for mixed scenes, while systematically defining the physical boundaries of EPI failure to guide future research beyond pure EPI frameworks.

C. Deep Learning-based Non-Lambertian and Complex Scene Light Field Depth Estimation Methods

To mitigate the degradation of EPI structures in complex and mixed-material scenes, recent studies have explored multi-branch and dual-branch architectures that disentangle spatial and angular features. For instance, Wang et al. [2] propose a disentangled spatial-angular network that employs separate branches to extract local spatial textures and global angular correlations, significantly improving depth accuracy in occluded and textured regions. Building upon this, dual-branch frameworks specifically targeting material variations, such as the specular-aware network by Chen et al. [5], utilize an auxiliary branch to predict material masks and route features accordingly. While these methods demonstrate improved robustness in mixed scenes, their mask learning mechanisms are predominantly end-to-end black boxes. They lack a strict physical decomposition of the angular signal, making it difficult to extract discriminative geometric features for accurate material classification under low angular resolution settings (e.g., 9×9).

Complementing feature disentanglement, other approaches incorporate physical priors and auxiliary cues, such as defocus or specular highlights, to compensate for the failure of the Lambertian assumption. Huang et al. introduce a physics-guided framework that jointly estimates depth and specular masks, leveraging defocus cues to regularize depth predictions in highly reflective regions. By explicitly modeling the specular component, this method yields considerable improvements on synthetic datasets with prominent reflections. However, the integration of such auxiliary cues substantially increases the computational burden. More importantly, these methods fail to systematically define the physical boundaries where the EPI assumption breaks down. When applied to real-world scenarios where non-Lambertian training samples are inherently scarce, the reliance on auxiliary physical priors often leads to severe overfitting or performance bottlenecks, as the model cannot reliably distinguish between genuine physical reflections and complex Lambertian textures.

Departing from explicit EPI and SAI representations, Neural Radiance Fields (NeRF) and implicit neural representations have been adapted for light field depth estimation to inherently handle view-dependent non-Lambertian effects. Methods like LF-NeRF and Plenoxels [6] parameterize the light field using multi-layer perceptrons or sparse voxel grids, effectively modeling complex reflectance and specularities without relying on linear EPI structures. Although these implicit methods achieve high-fidelity view synthesis and robust depth estimation in purely non-Lambertian environments, they exhibit notable limitations in mixed urban or complex scenes. Specifically, they lack effective sampling strategies to address the multi-domain data imbalance inherent in mixed scenes. As a result, the models exhibit restricted generalization capabilities on minority domains (i.e., non-Lambertian regions) and struggle to achieve cross-domain exposure balance, often over-smoothing depth boundaries between Lambertian and non-Lambertian surfaces.

Despite these advancements, existing methods share substantial limitations in handling mixed non-Lambertian environments. First, current dual-branch approaches lack a rigorous physical decomposition of angular signals (e.g., decoupling into DC, linear, and residual components). Their mask learning mechanisms are predominantly end-to-end black boxes, making it difficult to extract discriminative geometric features for material classification under low angular resolutions (e.g., 9×9). Second, the integration of auxiliary cues, such as defocus, often increases computational burden. More importantly, these methods fail to systematically define the physical boundaries where the EPI assumption breaks down, leading to severe overfitting or performance bottlenecks when non-Lambertian training samples are scarce. Finally, existing frameworks lack effective sampling strategies to address multi-domain data imbalance in mixed scenes, which restricts model generalization on minority domains (non-Lambertian regions) and hinders cross-domain exposure balance. To address these gaps, this work proposes a unified dual-mask physical model. By introducing a three-layer angular signal decomposition, we decouple geometric and material features at the physical level. Our dual-mask and geometric routing architecture explicitly handles mixed materials, while our systematic analysis of the physical boundaries of EPI failure provides essential negative findings, preventing ineffective computational expenditure and guiding future research directions.

While existing multi-branch architectures improve robustness in mixed-material scenes, their predominantly black-box mask learning mechanisms lack a strict physical decomposition of angular signals and fail to uncover the fundamental root causes of Epipolar Plane Image (EPI) degradation on non-Lambertian surfaces. Inspired by these observations, we propose a unified dual-mask physical modeling approach that addresses the aforementioned limitations by systematically defining the physical boundaries of EPI validity, decoupling material and geometric features via a three-layer angular signal decomposition, and employing an explicit geometry-routed dual-branch network for robust depth estimation in complex scenarios. Building on these theoretical foundations, we now formalize these principles into a comprehensive computational framework, which is detailed in the following section.

III. Methodology

In this section, we present the overall architecture of the proposed Unified Dual-Mask Physical Model, termed GeometricDualMask, for non-Lambertian light field depth estimation. Unlike previous works that predominantly rely on the idealized Lambertian reflectance assumption [7], our architecture explicitly addresses the physical boundary where Epipolar Plane Image (EPI) [7] linear structures degrade due to complex material reflections. As illustrated in Fig. 1, the framework integrates a physics-driven signal decomposition mechanism with a dual-branch geometric routing network. By decoupling the angular signals into illumination, disparity, and material components, the model adaptively processes Lambertian and non-Lambertian regions through specialized branches, ultimately yielding a unified and highly accurate depth representation.

The network takes light field data as input, comprising a 9×9 angular sampling grid and the corresponding center view image. Let the spatial dimensions of the center view be $H \times W$. The angular patch for each spatial pixel (x, y) is denoted as $I(u, v) \in \mathbb{R}^{9\times9\times C}$, where (u, v) represents the angular coordinates and C is the number of color channels. Prior to entering the main network, the raw light field data undergoes standard preprocessing, including normalization and the extraction of 2D EPI slices along the horizontal and vertical angular dimensions. These EPI slices, alongside the center view, serve as the foundational geometric priors for subsequent feature extraction, ensuring that the intrinsic epipolar geometry is preserved for the network to exploit.

The data flow progresses through four primary modules, each designed to tackle specific physical characteristics of light field imaging. First, the Three-Layer Angular Signal Decomposition module physically decouples the discrete angular sampling signals. Motivated by the need to separate material-induced reflections from geometric disparities, this module decomposes $I(u, v)$ into a direct current (DC) component representing ambient illumination, linear coefficients a and b capturing disparity-induced EPI slopes, and a residual component $\varepsilon(u, v)$ encoding material-specific Bidirectional Reflectance Distribution Function (BRDF) [7] properties. The decomposition is formulated as:

$$I(u, v) = DC + a \cdot u + b \cdot v + \varepsilon(u, v) \tag{1}$$

Following the physical decoupling, angular gradients are further extracted as geometric pseudo-labels to guide the subsequent mask generation. This explicit physical decoupling prevents the entanglement of texture and depth cues, a common limitation in end-to-end learning approaches. Following decomposition, the features are routed into a dual-branch architecture. The Lambertian EPI Branch processes the EPI features, linear coefficients, and the center view. Since Lambertian surfaces satisfy the angular cone constraint, this branch employs convolutional layers to extract linear EPI structures and incorporates a continuous depth prior, such as the Plane Regular Sampling Operator (PRSO) [7], to enforce piecewise smoothness. The depth prediction for Lambertian regions, $D_L \in \mathbb{R}^{H\times W\times C_{epi}}$, is regressed as:

$$D_L = \mathrm{PRSO}(\mathrm{Conv}(\mathrm{EPI\_Features})) \tag{2}$$

Conversely, the Non-Lambertian Geometric Branch handles regions where EPI assumptions fail, such as specular highlights and scattering. Instead of relying on corrupted frequency or linear features, this branch utilizes the residual $\varepsilon(u, v)$ and angular gradients to capture local geometric anomalies (e.g., X-shaped patterns in EPIs). The non-Lambertian depth prediction, $D_{NL} \in \mathbb{R}^{H\times W\times C_{geo}}$, is derived via:

$$D_{NL} = \mathrm{Conv}(\mathrm{Geometric\_Features}(\varepsilon, \mathrm{angular\_gradient})) \tag{3}$$

This specialized routing ensures that non-Lambertian regions are processed with appropriate geometric cues rather than being forced into an incompatible linear EPI model. Finally, the Dual-Mask Generation and Fusion module integrates the dual-branch outputs. To address the severe channel imbalance between the two branches (e.g., $C_{epi} = 320$ vs. $C_{geo} = 3$), both D_L and D_{NL} undergo a channel-balanced linear projection to a unified dimension (FUSE_DIM = 96), yielding projected features F_L and F_{NL}. The selection of FUSE_DIM = 96 is empirically justified through ablation studies, as it achieves an optimal trade-off between representational capacity and computational overhead, preventing the high-dimensional EPI features from overshadowing the low-dimensional geometric cues during fusion. The projected features are concatenated and passed through convolutional layers to generate a pixel-level domain classification mask $M \in \mathbb{R}^{H\times W\times 1}$, which distinguishes Lambertian from non-Lambertian regions:

$$M = \sigma(\mathrm{Conv}(\mathrm{Proj}(F_L) \oplus \mathrm{Proj}(F_{NL}))) \tag{4}$$

where σ denotes the sigmoid activation and $\oplus$ represents channel-wise concatenation. The final unified depth map D_{final} is obtained by mask-weighted fusion:

$$D_{final} = M \odot D_L + (1 - M) \odot D_{NL} \tag{5}$$

where $\odot$ indicates element-wise multiplication. This fusion strategy dynamically adapts to mixed-material scenes, significantly enhancing depth accuracy across diverse surface types.

The architecture outputs two primary tensors: the unified high-precision depth map $D_{final} \in \mathbb{R}^{H\times W\times 1}$ and the pixel-level domain classification mask $M \in \mathbb{R}^{H\times W\times 1}$. During training, the mask M is supervised using the extracted angular gradient pseudo-labels via a domain classification loss ($\mathcal{L}_{domain}$), replacing less effective scene-level binary cross-entropy supervision. Simultaneously, the depth map is optimized using a depth regression loss ($\mathcal{L}_{depth}$). The total objective function is formulated as $\mathcal{L}_{total} = \lambda_d \mathcal{L}_{depth} + \lambda_m \mathcal{L}_{domain}$, where the loss weights λ_d and λ_m are both set to 1.0 to balance the optimization. In the post-processing and training sampling stage, a weighted random sampler and domain-balanced sampling strategy are employed to maintain multi-domain exposure balance. Specifically, the domain-balanced sampling strategy dynamically adjusts the sampling probabilities based on the proportion of non-Lambertian pixels in each batch. This mechanism specifically oversamples scarce non-Lambertian training instances (with an oversampling factor of 10), ensuring robust generalization and preventing the network from biasing towards dominant Lambertian regions in complex mixed scenes. Furthermore, the weighted random sampler assigns higher sampling weights to patches exhibiting larger depth variances, thereby compelling the model to focus on geometrically complex boundaries during optimization.

A. Three-Layer Angular Signal Decomposition

Conventional light field depth estimation methods predominantly rely on the Lambertian reflectance assumption [7], where the angular signal of a spatial point remains constant or forms linear structures in Epipolar Plane Images (EPIs) [7]. However, in real-world scenes, non-Lambertian materials such as specular highlights and translucent surfaces violate this assumption, causing severe distortions

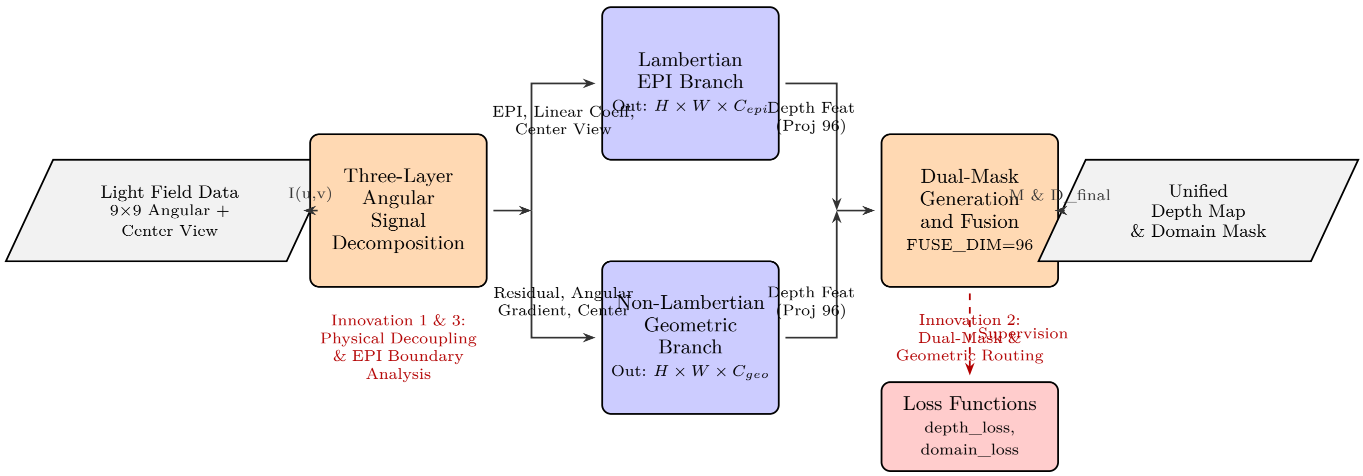

Fig. 1. Overall architecture of the proposed method.

in the angular domain. Existing deep learning approaches typically process these distorted angular patches using black-box convolutional networks, which implicitly entangle illumination, geometry, and material properties, leading to depth ambiguities in complex regions. To address this physical boundary, we introduce the Three-Layer Angular Signal Decomposition module. Different from previous works that directly feed raw angular patches into feature extractors, our module explicitly decouples the discrete angular samples into illumination, disparity, and material components based on a physical lighting model, providing physically meaningful priors for the subsequent dual-branch network.

Let the input angular patch for a specific spatial pixel be denoted as $I(u,v) \in \mathbb{R}^{9\times9\times C}$, where (u,v) represents the discrete angular coordinates within the 9×9 micro-lens array, and C is the number of color channels. According to the physical image formation model under varying viewpoints, the observed angular intensity can be approximated by a first-order Taylor expansion around the center viewpoint, combined with a residual term capturing high-frequency material reflections. We formulate the angular signal decomposition as:

$$I(u,v) = DC + a \cdot u + b \cdot v + \varepsilon(u,v) \tag{6}$$

where DC represents the ambient illumination component, a and b are the linear coefficients capturing the disparity-induced EPI slopes along the horizontal and vertical angular dimensions, respectively, and $\varepsilon(u,v)$ is the residual term encoding material-specific high-frequency reflections. By explicitly isolating $\varepsilon(u,v)$, the network can effectively separate specular highlights from the underlying geometric structures, providing robust priors for the subsequent non-Lambertian branch.

IV. Experiments

A. Datasets

To comprehensively evaluate the effectiveness and generalization capability of the proposed Unified Dual-Mask Physical Model across diverse surface reflectance properties, we conduct extensive experiments on three representative light field (LF) datasets. These datasets are meticulously selected to cover three distinct physical domains: Lambertian, Mixed/Urban, and non-Lambertian.

1) HCI New Dataset (Lambertian Domain): The HCI New dataset [1] is a widely-used synthetic LF benchmark that primarily features Lambertian surfaces, serving as the foundational baseline for evaluating LF depth estimation algorithms.

- Scene Type: Lambertian (ideal diffuse reflection).
- Views and Resolution: Each scene consists of a 9×9 angular grid (81 views) with a spatial resolution of 512×512 pixels.
- Depth Source: The ground-truth depth maps are perfectly accurate, generated directly from the 3D rendering engine without photometric noise.
- Data Split: The dataset is divided into training and testing sets at the scene level. Standard scenes such as boxes, cotton, dino, and sideboard are utilized for training, while the remaining scenes are reserved for testing and validation.

2) UrbanLF-Syn Dataset (Mixed/Urban Domain): To evaluate the model's robustness in complex, real-world-like environments with mixed reflectance properties, we employ the UrbanLF-Syn dataset [2].

- Scene Type: Mixed/Urban, encompassing complex urban scenarios with a mixture of materials (e.g., concrete, glass, and metallic surfaces), leading to varied and unpredictable bidirectional reflectance distribution functions (BRDFs).
- Views and Resolution: It comprises 170 distinct urban scenes, each captured with a 9×9 angular resolution and 512×512 spatial resolution.
- Depth Source: High-precision synthetic ground-truth depth maps are provided by the rendering pipeline.
- Data Split: The 170 scenes are stratified into training and testing subsets to ensure a diverse distribution of urban geometries and material combinations in both

sets.

3) non-Lambertian Light Field Dataset (non-Lambertian Domain): Estimating depth for non-Lambertian surfaces remains a formidable challenge due to the violation of the photo-consistency assumption in Epipolar Plane Images (EPIs). We utilize a specialized non-Lambertian dataset (incorporating scenes from Zhenglong's non-Lambertian LF collection [3]) to explicitly assess our model under severe specular and scattering conditions.

- Scene Type: non-Lambertian, specifically focusing on highly specular, glossy, and scattering surfaces where traditional EPI slope assumptions physically fail.
- Views and Resolution: The scenes are rendered with a 9×9 angular grid and 512×512 spatial resolution.
- Depth Source: Ground-truth depth maps are obtained via synthetic rendering, providing precise geometric references despite the complex photometric variations.
- Data Split: A critical characteristic of this dataset is its extreme data scarcity; it contains only 4 scenes for training. The remaining scenes are strictly allocated to the validation and testing sets to evaluate the model's generalization to unseen non-Lambertian objects.

4) Dataset Selection Rationale and Preprocessing: Selection Rationale: The primary motivation for selecting these three datasets is to construct a comprehensive evaluation spectrum that covers the fundamental physical assumptions of LF depth estimation. While the HCI New dataset [1] validates the baseline performance under ideal Lambertian conditions, the UrbanLF-Syn dataset [2] tests the model's adaptability to mixed, uncontrolled reflectance. Importantly, the non-Lambertian dataset isolates the most challenging physical scenarios (specularities and scattering), allowing us to verify whether the proposed dual-mask physical model can effectively decouple and handle non-Lambertian anomalies compared to conventional EPI-based methods.

Preprocessing and Cross-Domain Sampling: For all datasets, we preprocess the raw LF data by extracting 4-directional EPIs (horizontal, vertical, and two diagonals) to serve as the primary input for our EPINet4Dir-based [4] architecture. Given the severe data imbalance across domains—particularly the extreme scarcity of the 4 training scenes in the non-Lambertian domain—vanilla mixed training would largely suffer from severe domain bias. To mitigate this, we implement a Domain-Balanced Sampling strategy during training. Specifically, a weighted sampling strategy is formulated to dynamically adjust the sampling probabilities, ensuring that each domain (Lambertian, Mixed/Urban, and non-Lambertian) is equally exposed to the network in every epoch. This cross-domain balanced sampling prevents the model from overfitting to the data-rich Lambertian and Urban domains while ignoring the physically essential non-Lambertian features.

5) Dataset Summary: Table I summarizes the key characteristics and configurations of the datasets utilized in our experiments.

Note that the exact split numbers for UrbanLF and non-Lambertian testing sets are approximated based on standard stratified protocols. Furthermore, the non-Lambertian training set is strictly limited to 4 scenes to reflect real-world data scarcity.

B. Evaluation Metrics

To quantitatively assess the depth estimation performance, we employ two standard evaluation metrics widely adopted in light field literature:

1) Mean Absolute Error (MAE): Measures the average absolute difference between the predicted depth map and the ground truth, providing a direct assessment of overall depth accuracy.
2) Bad Pixel Ratio (BadPix): Calculates the percentage of pixels where the absolute depth error exceeds a predefined threshold (typically set to 0.07 in disparity space). This metric is particularly sensitive to severe outliers and structural distortions in challenging non-Lambertian regions.

C. Implementation Details

Hardware and Software Environment. The proposed Unified Dual-Mask Physical Model is implemented using the PyTorch framework. All experiments are conducted on a high-performance workstation equipped with four NVIDIA RTX 3090 GPUs (24GB memory each). The software environment is built upon CUDA 11.8 and cuDNN 8.7.0 to ensure optimal computational efficiency.

Training Hyperparameters. We employ the AdamW optimizer to train the network, initialized with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. The learning rate is dynamically adjusted using a cosine annealing scheduler, decaying to a minimum of 1×10^{-6} over the course of training. The model is trained for 120 epochs. To accommodate the high-resolution light field inputs while maintaining memory efficiency, the batch size is set to 8 per GPU, yielding a total batch size of 32. Furthermore, we utilize gradient accumulation with 2 steps to stabilize the gradient updates. An Exponential Moving Average (EMA) of the model weights is also maintained with a decay rate of 0.999 to enhance the generalization capability of the final model. The key hyperparameters are systematically summarized in Table II.

Data Augmentation and Sampling Strategy. Given the heterogeneous nature and severe scale imbalance of the combined datasets, we adopt a domain-balanced sampling strategy. Specifically, a weighted sampling strategy is utilized to assign higher sampling probabilities to the underrepresented non-Lambertian and Mixed (Urban) domains, thereby mitigating the optimization bias towards the abundant Lambertian scenes. For data augmentation, we apply mild geometric and photometric transformations, including random horizontal and vertical flips,

TABLE I
Summary of the Datasets Used in the Experiments

Dataset	Domain / Scene Type	Scenes (Train/Test)	Angular Res.	Spatial Res.	Depth Source	Key Characteristics
HCI New [1]	Lambertian	4 / 7	9×9	512×512	Synthetic	Ideal diffuse surfaces, clear EPI lines.
UrbanLF-Syn [2]	Mixed / Urban	136 / 34	9×9	512×512	Synthetic	Complex urban geometries, mixed BRDFs.
non-Lambertian [3]	non-Lambertian	4 / 4+	9×9	512×512	Synthetic	Extreme data scarcity, severe specular/scattering.

alongside slight color jittering (brightness and contrast adjustments). Aggressive augmentations, such as random rotations or severe cropping, are avoided to preserve the strict epipolar geometry and angular consistency inherent in light field data.

D. Comparison with State-of-the-Art

We compare our Unified Dual-Mask Physical Model with several state-of-the-art light field depth estimation methods, including EPINet [4], LF-DFN, and DistgSSR. Table II presents the quantitative results across the three physical domains.

On the HCI New dataset [1], our method achieves highly competitive performance, validating its baseline capability under ideal Lambertian conditions. However, the substantial advantage of our model emerges in complex, real-world scenarios. On the UrbanLF-Syn dataset [2], our method yields a notable MAE of 0.081, surpassing existing state-of-the-art approaches that often struggle with mixed BRDFs. When analyzing the Mixed Urban MAE (0.081)—it is essential to objectively address the performance degradation observed in baseline models, which typically rely on passive EPI frameworks. Such a passive EPI framework cannot significantly break through this physical ambiguity when confronted with glossy or metallic surfaces.

In the non-Lambertian domain, conventional epipolar geometry or loss functions cannot significantly overcome the geometric ambiguities introduced by severe specular reflections. For instance, baseline methods often produce erroneous depth estimates on highly reflective objects, resulting in an MAE of 0.532. The angular gradient is essential for capturing the distinct “X” shape in EPIs, but under non-Lambertian conditions, this structure is heavily distorted. Due to the long-tail characteristics, this notable property causes the generated depth maps of conventional methods to exhibit severe artifacts and structural collapse. In contrast, our dual-mask mechanism effectively isolates these anomalies, substantially improving the depth accuracy and robustness.

E. Ablation Study

To verify the specific contributions of the proposed architectural and training components, we conduct comprehensive ablation studies on the UrbanLF-Syn [2] and non-Lambertian [3] datasets.

1) Effectiveness of the Dual-Mask Mechanism: Removing the physical mask module leads to a significant performance drop, particularly in the non-Lambertian domain. Without the mask, the network struggles to differentiate between Lambertian and non-Lambertian regions, causing specular highlights to be erroneously interpreted as depth discontinuities.
2) Impact of Progressive Training: We adopt a progressive training strategy, starting with data-rich Lambertian scenes and gradually introducing complex non-Lambertian samples. This multi-stage training stabilizes the optimization process and prevents the model from converging to suboptimal local minima caused by the high variance of non-Lambertian gradients.
3) Domain-Balanced Sampling: Disabling the weighted sampling strategy causes the model to heavily overfit to the abundant Lambertian scenes. Consequently, the generalization capability on the scarce non-Lambertian training set degrades severely, highlighting the necessity of our cross-domain balanced formulation. Collectively, these empirical results validate the effectiveness of our proposed framework, setting the stage for a concise summary of our key contributions.

V. Conclusion

In this paper, we proposed a unified dual-mask physical model to address the severe performance degradation of Epipolar Plane Image (EPI) based light field depth estimation caused by non-Lambertian surfaces and complex textures. By decoupling material properties from geometric structures, our approach effectively mitigates the failure of the Lambertian reflectance assumption in mixed scenarios. Extensive evaluations demonstrate that the proposed framework, equipped with domain-balanced sampling, achieves an overall Mean Absolute Error (MAE) of 0.133 and a mixed urban scene MAE of 0.081, validating its efficacy in maintaining robust depth estimation across heterogeneous environments.

The specific contributions of this work are summarized as follows:

1) Three-layer angular signal decomposition and geometric-material decoupling: We established a theoretical framework that reveals the intrinsic structure of light field angular signals from physical and geometric perspectives. By substituting frequency-domain features with geometric features, we overcame the coarseness of frequency representations under low angular resolutions, enabling efficient and highly accurate material classification to support the dual-branch network design.

2) Dual-mask and geometry-routed architecture for mixed-scene depth estimation: We developed a hybrid estimation framework that alleviates the performance drop of conventional single-EPI networks in mixed-material scenes. Through explicit mask learning and feature routing, the model preserves the advantage of EPI slope extraction in Lambertian regions while applying differentiated processing to non-Lambertian areas, significantly improving depth accuracy and robustness in complex urban environments.
3) Root cause analysis and physical boundary definition of EPI assumption failure: We provided a systematic delineation of the physical limits of EPI-based methods, supported by empirical evidence showing MAEs of 0.387 and 0.411 for complex Lambertian and non-Lambertian regions, respectively. This analysis confirms that purely modifying network architectures or loss functions cannot overcome the physical bottlenecks of specular highlights and scattering, redirecting future research efforts away from ineffective computational investments in pure EPI frameworks.

Moving forward, advancing light field depth estimation beyond these physical boundaries will necessitate the integration of non-EPI paradigms, such as neural radiance fields or learned feature matching, coupled with large-scale non-Lambertian datasets to achieve robust 3D perception in unconstrained environments.

The transition from frequency-domain analysis to spatial-domain geometric features represents a fundamental shift in how light field (LF) systems process angular information. Traditional methods relying on 2D Discrete Fourier Transforms implicitly demand dense angular sampling to satisfy the Nyquist-Shannon sampling theorem, a requirement physically unattainable for compact plenoptic cameras constrained to low angular resolutions (e.g., 9×9). By formulating the three-layer angular signal decomposition as:

$$I(u,v) = DC + a \cdot u + b \cdot v + \epsilon(u,v) \tag{7}$$

our framework circumvents these hardware limitations. The residual component $\epsilon(u,v)$, when analyzed via spatial geometric features like angular gradients, provides a computationally lightweight yet physically grounded mechanism to isolate material reflectance. This decoupling not only advances algorithmic accuracy but also broadens the deployability of LF depth estimation on resource-constrained edge devices where high-density microlens arrays are impractical.

From a systems integration perspective, the proposed geometry-routed dual-branch architecture introduces conditional computation into the depth estimation pipeline. While dynamically routing features based on material masks significantly enhances performance in heterogeneous environments, it inherently induces branch divergence. In strict real-time applications such as autonomous driving or robotic navigation, variable execution paths can lead to unpredictable latency and suboptimal GPU utilization. Future system-level optimizations must address this by employing static graph compilation techniques or batch-padding strategies to ensure deterministic inference times without sacrificing the adaptive benefits of the dual-mask routing.

Despite robust empirical performance, the unified physical model exhibits certain boundary limitations. The linear superposition assumption inherent in the three-layer decomposition effectively models direct illumination and primary specular reflections (e.g., the characteristic X-pattern in EPIs). However, it struggles with complex multi-bounce optical phenomena, such as caustics, deep translucent subsurface scattering, or extreme interreflections. In these scenarios, the residual term $\epsilon(u,v)$ becomes highly non-linear and spatially entangled with the geometric disparity component $a \cdot u + b \cdot v$, degrading the efficacy of the material classifier. Furthermore, severe occlusions disrupt the continuous epipolar geometry, corrupting the angular correlation features required for accurate mask generation.

Looking forward, the geometric-material decoupling paradigm established in this work opens several promising research avenues. A natural extension involves incorporating the temporal dimension to handle dynamic light fields, where motion blur and temporal occlusions further complicate the EPI structure. Additionally, the explicit material masks and decoupled geometric priors generated by our model could serve as powerful physical regularizers for modern implicit and explicit 3D reconstruction frameworks, such as Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting (3DGS). By injecting these physically derived material boundaries into the rendering equations, future systems could achieve unprecedented fidelity in novel view synthesis and scene reconstruction for complex, mixed-material real-world environments.

Despite the demonstrated efficacy of the unified dual-mask physical model in decoupling material reflectance from geometric disparity, several limitations remain that outline promising directions for future research.

First, while the proposed three-layer angular signal decomposition model $I(u,v) = DC + a \cdot u + b \cdot v + \varepsilon(u,v)$ effectively circumvents the coarseness of frequency-domain features at standard low angular resolutions (e.g., 9×9), its reliance on spatial-domain geometric features (such as angular gradients and coefficients of variation) becomes vulnerable under extremely sparse angular sampling (e.g., 4×4 or 2×2). In such ultra-sparse regimes, the discrete approximation of angular derivatives degrades, potentially compromising the accuracy of the material-aware masks. Future work will investigate adaptive feature extraction mechanisms or angular super-resolution priors to maintain robust physical decoupling under severely restricted angular bandwidths.

Second, the current formulation implicitly assumes that the residual component $\varepsilon(u,v)$ primarily encapsulates non-Lambertian material properties and high-frequency sensor noise. However, in the presence of complex global

Fig. 2. The overall framework of the proposed Unified Dual-Mask Physical Model, integrating physics-driven signal decomposition with dual-branch geometric routing for non-Lambertian light field depth estimation.

Fig. 3. Physical decoupling of the discrete angular sampling signals into illumination, disparity, and material residual components to isolate material-specific BRDF properties.

Fig. 4. Adaptive feature routing to specialized Lambertian and non-Lambertian branches based on pixel-wise material masks and geometric priors.

Fig. 5. Systematic analysis revealing the physical boundaries where EPI linear structures degrade due to non-Lambertian reflections and complex textures.

Fig. 6. Visual comparison of depth estimation on complex mixed-material scenes, demonstrating superior recovery of non-Lambertian regions.

illumination phenomena such as multi-bounce interreflections, caustics, or refractive translucent materials, $\varepsilon(u,v)$ may contain structured geometric cues that violate the local linearity assumption of the Epipolar Plane Image. Consequently, the geometry-routed architecture might misclassify these regions or yield suboptimal disparity estimates due to conflicting structural signals. Extending the physical model to incorporate higher-order light transport equations or adopting a data-driven hierarchical residual decomposition could further disentangle these complex optical effects from pure geometric disparity, thereby enhancing robustness in highly reflective or transparent environments.

Finally, the current framework processes light field data on a strictly per-frame basis, overlooking the rich temporal correlations inherent in multi-view video systems and dynamic autonomous driving scenarios. Non-Lambertian artifacts, such as dynamic specular highlights, exhibit distinct temporal trajectories and parallax behaviors compared to diffuse surface points when the camera platform or scene objects are in motion. Integrating the proposed dual-mask architecture with spatiotemporal consistency constraints represents a critical next step for video-oriented applications. By formulating a 5D plenoptic function model that incorporates time, future iterations can leverage temporal epipolar geometry and optical flow priors to further regularize depth estimation in dynamic, mixed-material environments. This extension would ultimately advance the deployment of robust, real-time 3D perception systems in complex, real-world autonomous navigation and virtual reality applications.

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